



Summarizing the results of data quality tests

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&

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Felix Eckel

Luděk Vecsey

AdriaArray Workshop, March 12-15, 2024, Sofia
The workshop is partly supported by ORFEUS



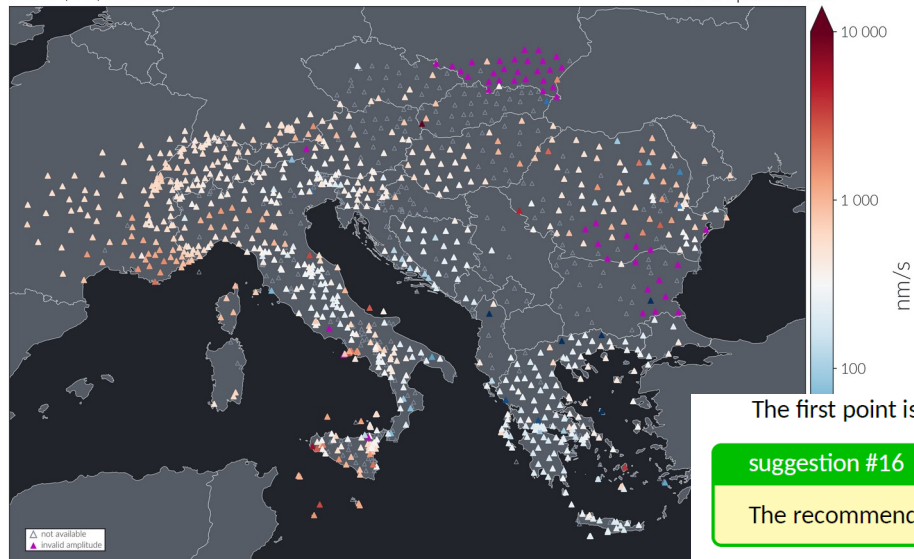
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UAG 2023 report – suggestions – metadata handling

5.0 s | Z | 2023-03-03

noise levels
95th percentile



inconsistency in handling the StationXML files by different version of SeisComP, ObsPy, (not)following the FDSN StationXML schema

The first point is that different EIDA nodes treat the metadata differently.

suggestion #16

The recommendation would be to use the same version of SeisComP at all EIDA nodes.

However, we need to note that the error was only revealed right thanks to the fact that the versions are different, and hence the different behavior could have been tracked.

suggestion #17

The more important recommendation would hence be, that the same StationXML scheme is followed by all the software developers (SeisComp, ObsPy, ...).

There is only one valid StationXML scheme, which is the one by FDSN <https://docs.fdsn.org/projects/stationxml/en/latest/>.

UAG 2023 report – suggestions - StationXML

suggestion #2

The UAG recommends ORFEUS/EIDA to promote a uniform way to produce StationXML, so that inexperienced users can be advised on the techniques and tools available to handle StationXML files.

suggestion #3

In addition, ORFEUS could organize webinar or workshop sessions devoted to the management of StationXML, where station operators can share tools and discuss their direct experiences. The AdriaArray project clearly shows that training in this direction is needed.

stationXML documentation

```
<FDSNStationXML schemaVersion="1.2">
  <Source>SeisCompP</Source>
  <Sender>GFZ</Sender>
  <Created>2023-04-02T17:49:15.915266Z</Created>
  <Network code="Z3" startDate="2015-01-01T00:00:00Z" endDate="2022-12-31T00:00:00Z" restrictedStatus="open">
    <Description>
      AlpArray Seismic Network (AASN) temporary component
    </Description>
    <Identifier type="DOI">10.12686/alparray/z3_2015</Identifier>
  </Network>
  <Station code="A412A" startDate="2017-06-15T16:45:38Z" endDate="2018-02-12T09:54:14Z" restrictedStatus="open">
    <Comment id="1">
```

revision of 1.1 ==> 1.2 by
Philip Crotwell
Robert Casey
Jonathan Schaeffer
Petr Kolínský
accepted in 2022 by FDSN

StationXML



v1.2

Search docs

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2. StationXML Reference

2.1. <FDSNStationXML> required

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2.2.2. <Identifier>

2.2.3. <Comment>

2.2.4. <DataAvailability>

2.2.5. <Operator>

2.2.6. <TotalNumberStations>

2.2.7. <SelectedNumberStations>

2.3. <Station>

2.4. <Channel>

2.5. <Response>

3. Specifying and Using Response Information

4. StationXML Tools

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Read the Docs

v1.2

2.2. <Network> required

Warning: Future Change

endDate: This attribute should not be used if it is in the future.

Warning: Future Change

endDate: This attribute is likely to require timezone of Z.

Warning: Future Change

startDate: This attribute is likely to require timezone of Z.

The Network container. All station metadata for this network is contained within this element. A Description element may be included with the official network name and other descriptive information. An Identifier element may be included to designate a persistent identifier (e.g. DOI) to use for citation. A Comment element may be included for additional comments.

An active Network should not use the endDate attribute. Unlike SEED, do not use an endDate in the distant future to mean active.

Example: <Network code="XX" startDate="2016-01-27T13:00:00Z" />

Attributes of <Network>:

attribute	type	required	description	example
alternateCode	string	no	A code used for display or association	alternateCode="GSN"
code	string	yes	Name of Network	code="XX"
endDate	dateTime	no	End date of network. Do not use if still active, endDate should not be in the future.	endDate="2018-01-27T00:00:00Z"
historicalCode	string	no	A previously used code if different from the current code	historicalCode="XX"
restrictedStatus	RestrictedStatusType	no	One of: "open", "closed", "partial"	restrictedStatus="open"
sourceID	anyURI	no	A data source identifier in URI form. It is recommended that this follow the FDSN Source Identifiers specification, http://docs.fdsn.org/projects/source-identifiers	sourceID="FDSN.XX"
startDate	dateTime	no	Start date of network	startDate="2016-07-01T00:00:00Z"

Sub Elements of <Network>

element	type	number
<Description>	string	optional

ORFEUS:

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Quality Assurance

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ORFEUS Software Development Grants

ORFEUS established a Software Development Fund in 2021. A Call for Applications opens every spring. Software grants are awarded for the development, consolidation, implementation of relevant software to ease the management of, access to, and processing of data and metadata integrated in the ORFEUS EIDA (European Integrated Data Archive) and Strong-Motion programs. This also includes: processing tools tailored to new quality metrics; software libraries that can be used centrally as well as by external clients; software useful to prepare the ORFEUS infrastructure to deal with new data types and communities.

Call 2021: <https://polybox.ethz.ch/index.php/s/hWq2ycAcBJcO5MUv>. The recipients of the 2021 ORFEUS Software Development Grants are as follows:

- Gesa Petersen (GFZ) - project *"Extension and implementation of AutoStatsQ for an updating assessment of the quality of seismic data and metadata of seismic networks"*
 - Webinar: <https://polybox.ethz.ch/index.php/s/DDd23LxpreJb2FH>
 - Repository: <https://github.com/gesape/AutoStatsQ>
- Peter Makus (GFZ) - project *"Pythonic Global Lithospheric Imaging using Earthquake Recordings (PyGLImER)"*
 - Webinar: <https://polybox.ethz.ch/index.php/s/mdsSGRINBkt7OPc>
 - Repository: <https://github.com/PyGLImER/PyGLImER>
- Mathijs Koymans (KNMI) - *"Integration of Continuous Terrestrial Gravimetric Data from the NEWTON-g Gravity Imager with ORFEUS/EIDA"*.
 - Poster at ORFEUS annula workshop 2022: <https://polybox.ethz.ch/index.php/s/53oBmOiiHyPamD5>
 - Repository: <https://github.com/NEWTON-g>

Call 2022: <https://polybox.ethz.ch/index.php/s/4LE7FgYRwwHpVxS>. The recipients of the 2022 ORFEUS Software Development Grants are as follows:

- Johannes Stampa (CAU-KIEL) - project *"Data Availability Monitoring Tools"*
- Claudia Mascandola (INGV-MI) - project *"ESMpro"*

EIDA:

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EPOS Seismology Workshop 2023

ADRIAARRAY INITIATIVE:

AdriaArray

AdriaArray - Organization

AdriaArray - Working Groups

AdriaArray - Data analysis and research

AdriaArray - Seismic networks

AdriaArray Communication & Outreach

ORFEUS Data Integration Grants

Data integration grants are provided by ORFEUS within the Horizon Europe project Geo-INQUIRE with the chief goals of: (a) enhancing seismological waveform data integration from regions with sparse data coverage in the EIDA (European Integrated Data Archive) or ESM (Engineering Strong Motion) database, and (b) representing new disciplines and types of Earth Science data crucial in the geo- INQUIRE framework (e.g. infrasound, GNSS, DAS, OBS, gravimetry, etc.). Grants will be awarded to support finalizing the permanent integration of fully open waveform datasets and associated station / site metadata. The institutions integrating data sets must provide clear evidence of ownership, open license (e.g. CC-BY 4.0 or equivalent) and detailed metadata in the formats handled by the EIDA and ESM archives. The grants are intended to support the finalization of data curation including QC to allow integration in ORFEUS-associated archives.

Call 2023: <https://polybox.ethz.ch/index.php/s/2gHQ1pzcEvnEw2P>.

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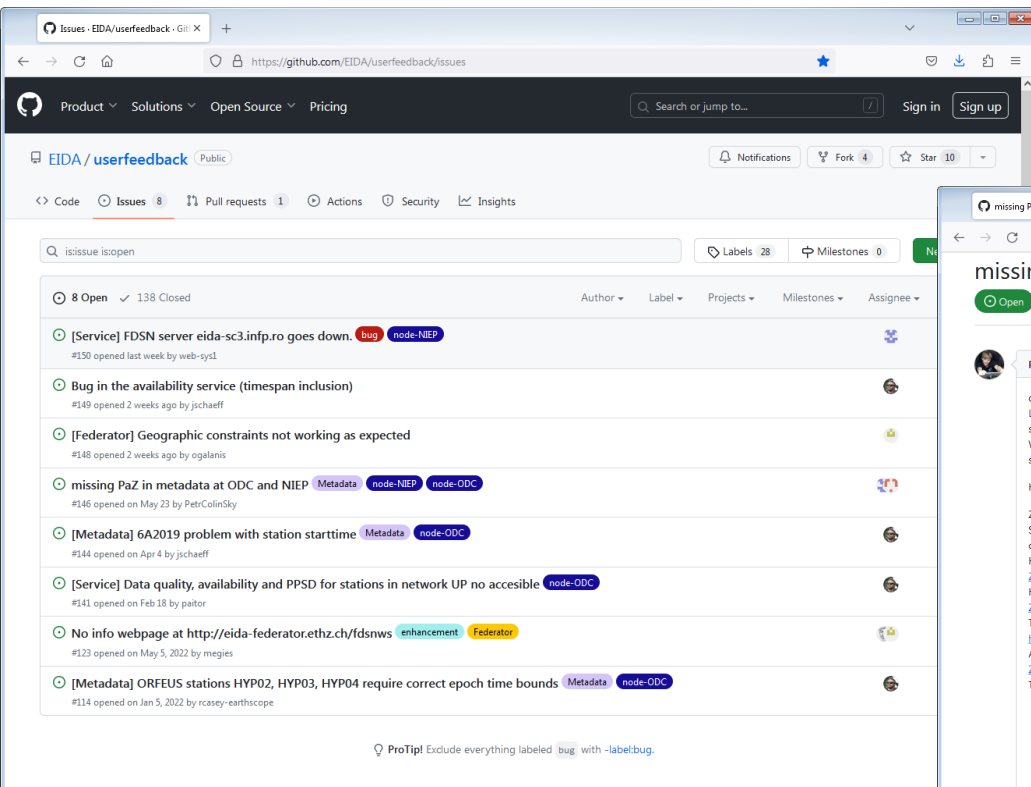
Ad by EthicalAds · 

© Copyright 2020, ORFEUS. Revision ffb673b1.

Built with [Sphinx](#) using a [theme](#) provided by [Read the Docs](#).

EIDA issue tracker

<https://github.com/EIDA/userfeedback/issues>



This screenshot shows the GitHub interface for the EIDA/userfeedback repository. The top navigation bar includes links for Product, Solutions, Open Source, and Pricing, along with a search bar and login/sign-up buttons. The repository name 'EIDA / userfeedback' is displayed with a 'Public' badge. Below this, tabs for Code, Issues (8), Pull requests (1), Actions, Security, and Insights are visible. A search bar contains the text 'is:issue is:open'. To the right, filters for Labels (28) and Milestones (0) are shown. The main content area lists eight open issues, each with a title, a brief description, the number of comments, the author, and the date opened. The issues are: 1. '[Service] FDSN server eida-sc3.infp.ro goes down.' (bug, node-NIEP), 2. 'Bug in the availability service (timespan inclusion)', 3. '[Fedorator] Geographic constraints not working as expected', 4. 'missing PaZ in metadata at ODC and NIEP' (Metadata, node-NIEP, node-ODC), 5. '[Metadata] 6A2019 problem with station starttime' (Metadata, node-ODC), 6. '[Service] Data quality, availability and PPSD for stations in network UP no accessible' (node-ODC), 7. 'No info webpage at http://eida-fedorator.ethz.ch/fdsnws' (enhancement, Fedorator), and 8. '[Metadata] ORFEUS stations HYP02, HYP03, HYP04 require correct epoch time bounds' (Metadata, node-ODC). A 'ProTip!' at the bottom suggests excluding everything labeled 'bug' with the filter '-label:bug'.

Issues · EIDA/userfeedback · Gill X

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is:issue is:open

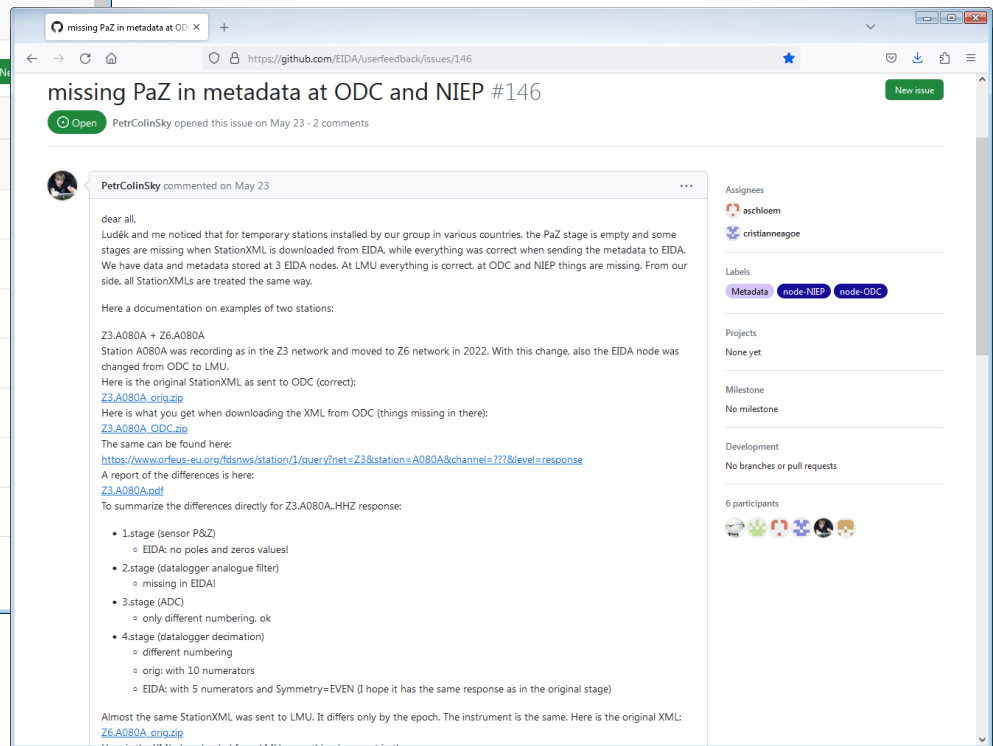
Labels 28 Milestones 0

8 Open 138 Closed

Author Label Projects Milestones Assignee

- [Service] FDSN server eida-sc3.infp.ro goes down. bug node-NIEP
#150 opened last week by web-sys1
- Bug in the availability service (timespan inclusion)
#149 opened 2 weeks ago by jschaeff
- [Fedorator] Geographic constraints not working as expected
#148 opened 2 weeks ago by ogalanis
- missing PaZ in metadata at ODC and NIEP Metadata node-NIEP node-ODC
#146 opened on May 23 by PetrColinSky
- [Metadata] 6A2019 problem with station starttime Metadata node-ODC
#144 opened on Apr 4 by jschaeff
- [Service] Data quality, availability and PPSD for stations in network UP no accessible node-ODC
#141 opened on Feb 18 by paitor
- No info webpage at http://eida-fedorator.ethz.ch/fdsnws enhancement Fedorator
#123 opened on May 5, 2022 by megies
- [Metadata] ORFEUS stations HYP02, HYP03, HYP04 require correct epoch time bounds Metadata node-ODC
#114 opened on Jan 5, 2022 by rcassey-earthscope

ProTip! Exclude everything labeled bug with -label:bug.



This screenshot shows the details of issue #146, 'missing PaZ in metadata at ODC and NIEP'. The issue is marked as 'Open' and was created by PetrColinSky on May 23, 2022, with 2 comments. The issue title is 'missing PaZ in metadata at ODC and NIEP #146'. The issue is labeled with 'Open', 'bug', 'node-NIEP', and 'node-ODC'. The issue description states: 'dear all, Luděk and me noticed that for temporary stations installed by our group in various countries, the PaZ stage is empty and some stages are missing when StationXML is downloaded from EIDA, while everything was correct when sending the metadata to EIDA. We have data and metadata stored at 3 EIDA nodes. At LMU everything is correct, at ODC and NIEP things are missing. From our side, all StationXMLs are treated the same way. Here a documentation on examples of two stations: Z3.A080A + Z6.A080A Station A080A was recording as in the Z3 network and moved to Z6 network in 2022. With this change, also the EIDA node was changed from ODC to LMU. Here is the original StationXML as sent to ODC (correct): Z3.A080A_orig.zip Here is what you get when downloading the XML from ODC (things missing in there): Z3.A080A_ODC.zip The same can be found here: https://www.orfeus-eu.org/fdsnws/station/?query=net=Z3&station=A080A&channel=?&level=response A report of the differences is here: Z3.A080A.pdf To summarize the differences directly for Z3.A080A.HHZ response: 1.stage (sensor P&Z) o EIDA: no poles and zeros values! 2.stage (datalogger analogue filter) o missing in EIDA! 3.stage (ADC) o only different numbering, ok 4.stage (datalogger decimation) o different numbering o orig: with 10 numerators o EIDA: with 5 numerators and Symmetry=EVEN (I hope it has the same response as in the original stage) Almost the same StationXML was sent to LMU. It differs only by the epoch. The instrument is the same. Here is the original XML: Z6.A080A_orig.zip'. The right sidebar shows the assignees (aschloem, cristianneage), labels (Metadata, node-NIEP, node-ODC), projects (None yet), milestones (None yet), and development status (No branches or pull requests). It also shows 6 participants.

missing PaZ in metadata at ODC and NIEP #146

Open PetrColinSky opened this issue on May 23 - 2 comments

PetrColinSky commented on May 23

dear all,
Luděk and me noticed that for temporary stations installed by our group in various countries, the PaZ stage is empty and some stages are missing when StationXML is downloaded from EIDA, while everything was correct when sending the metadata to EIDA. We have data and metadata stored at 3 EIDA nodes. At LMU everything is correct, at ODC and NIEP things are missing. From our side, all StationXMLs are treated the same way.
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To summarize the differences directly for Z3.A080A.HHZ response:

- 1.stage (sensor P&Z)
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[Z6.A080A_orig.zip](#)

Assignees
aschloem
cristianneage

Labels
Metadata node-NIEP node-ODC

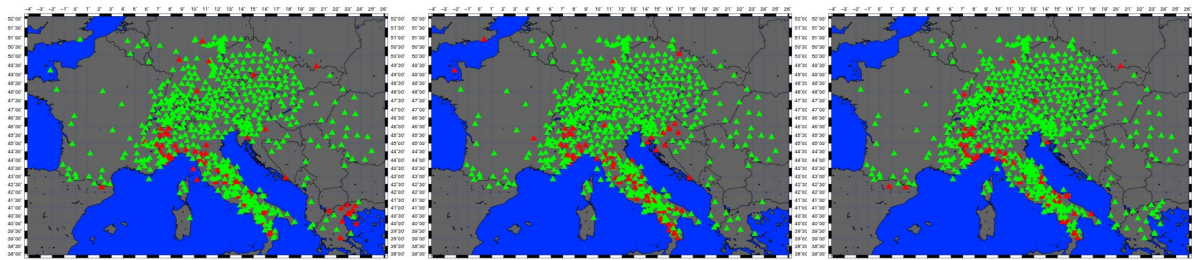
Projects
None yet

Milestone
None yet

Development
No branches or pull requests

6 participants

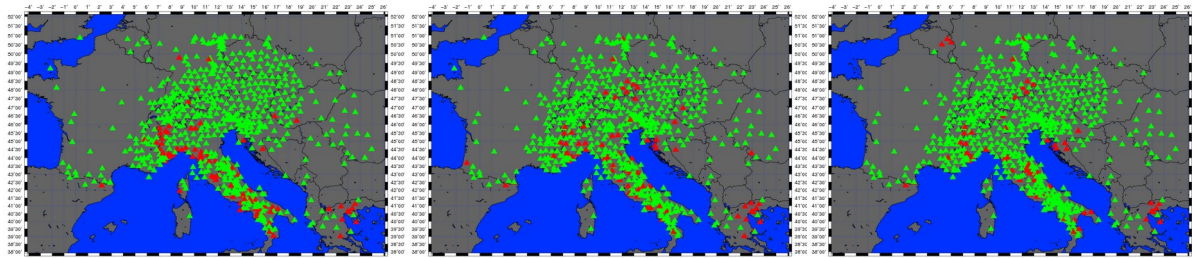
data Q - look back



Kamchatka M=7.2, 30 **January** 2016

Sumatra M=7.8, 2 **March** 2016

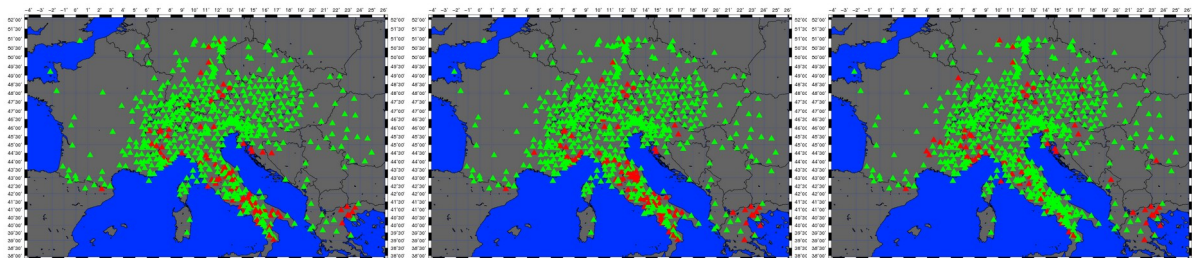
Morocco M=5.2, 15 **March** 2016



Ecuador M=7.8, 16 **April** 2016

South Atlantic M=7.4, 19 **August** 2016

Ascension M=7.1, 29 **August** 2016



Greece M=5.5, 15 **October** 2016

Italy M=6.5, 30 **October** 2016

New Zealand M=7.8, 13 **November** 2016

EGU **2017** talk by
Kolínský et al.
dataset: AlpArray

... one more test: **event-data oriented**

technical check for quality of data in terms of

- gaps
- overlaps
- missing traces
- gains of components
- spikes
- zero traces
- response removal
- ...

stage 1 – automatic check

- if there are right 3 traces with no gaps and no overlaps for a given station, the record is labeled **+**
 - if there are more than 3 traces with no gaps and no overlaps differing just by location code, we select 3 traces with the same code and the record is labeled as **selected**
 - if for any of the component, there are more than 3 traces but less than 20 traces, we check for the length of the gaps; if no gap is longer than 3 s, we merge (interpolate) the traces and we label the record as **merged**
 - the same apply also for overlaps, the record is labeled as **ovrlps mrgd**
 - if for any of the component, there are more than 20 traces, the record is labeled as **many trcs**
 - if for any of the component, there are less than 20 traces, but any of the gaps is longer than 3 sec, the record is labeled as **long gaps**
 - if any of the component is missing, the record is labeled as **missing**
 - if the record of any component is shorter than the required time window (usually 1.5 – 2.0 hrs), it is labeled as **short rec**
 - if the record is not available in the whole time window, it is labeled as **black**
- Greens** and **yellows** are used further, **reds** are discarded.

followed by visual inspection

stage 2 – visual inspection

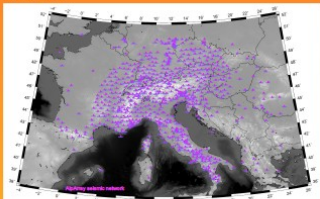
After the **greens** and **yellows** are selected, we plot the geometrically rotated (ZRT) records and inspect them visually to remove zero traces and records with electronic non-seismic noise. Records with non-visible earthquake record (noisy) and with significant peaks are also removed. These are then labeled as **elc noise**, **zeros**, **noisy** and **peaks** (with possible Z-R-T component affected). Since we plot the records with respect to the epicentral distance, we also roughly check for the record timing. If the record is significantly out of time, it is labeled as **time shift**. If traces for some components are equal, we label it as **R = T**. All the earthquakes were big events, so, if not visible at the record, something is wrong, especially when it is well visible at the neighboring stations. What still remains **green** and **yellow** is forwarded for the analysis.

Some of the permanent stations are **black** for all the earthquakes. These stations may be non-existent. The question is then, why they are listed as “open” at Orfeus.

Maps for the nine earthquakes after the **automatic** check are on the next page. **Green** in map means **green** + **yellow** + **purple** in the table. **Purples** were removed visually after the maps were plotted. **Red** in the map is **red** in the list.

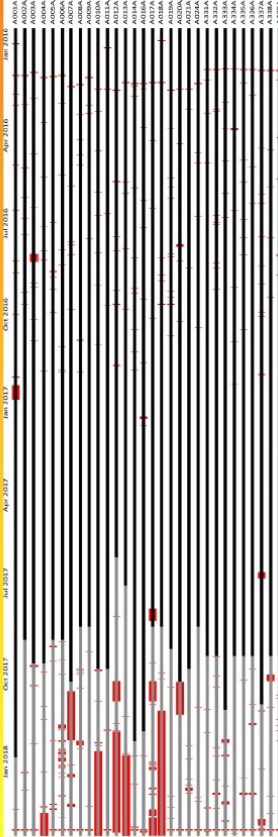
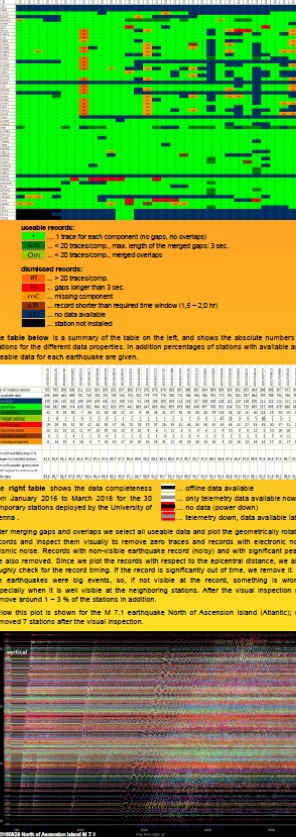
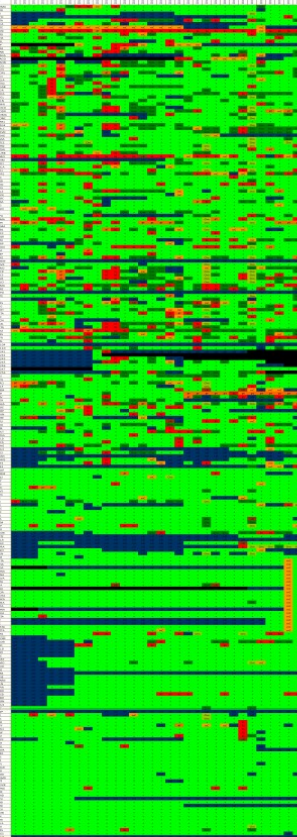
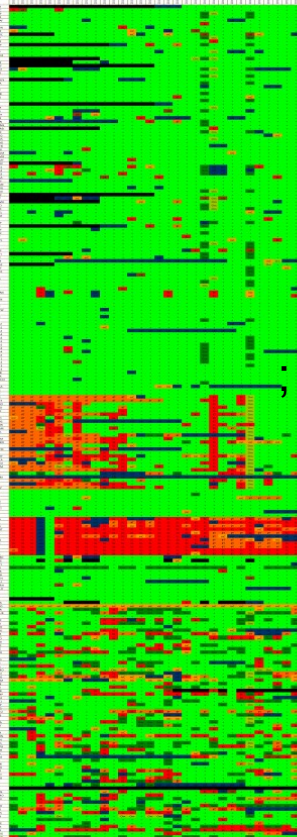
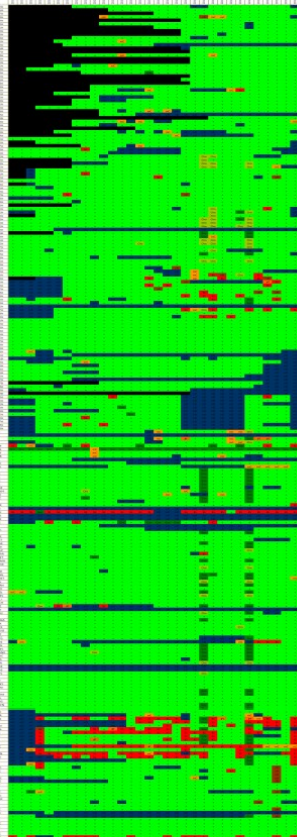
sharing the Q test results

The AlpArray research project allows us to investigate the structure and evolution of the lithosphere beneath the entire Alpine area. The project includes around 250 temporary seismic stations, which have been deployed in the period from spring 2016 to summer 2017. Together with the included permanent stations from all over Europe, there are around 500 stations available in total.



For the surface wave analysis, we are interested in two-hour time windows for each earthquake. For different reasons we cannot use the data of all stations for the data processing. These are the records that are not available for the given day, components are missing, there are too many gaps or overlaps, the traces are shifted in time, the records are too noisy or have no seismic signal. We selected 32 earthquakes spanning 2016 and 2017 and investigate the quality of the data before the surface wave analysis.

The following table contains information about availability and quality of the data for 918 stations and respective time windows for each earthquake.



how can you contribute

- so far, we have included tests by four „testers“
- additional tests are welcome

output of your test = input for the summarizing table

simple csv or json file

XY.NAME,value

- your test needs to cover a significant portion of stations (tens? hundreds?)
- you provide a description of the test (half-a-page)
- you provide detailed results if applicable (accessible online)
- you suggest color coding for your output values

percentage ... numbers ... flags ... True/False ... keywords ... special characters



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Orfeus



acknowledgements

Maps plotted using GMT by Wessel, P., Smith, W. H. F., Scharroo, R., Luis, J. F. and Wobbe, F., 2013: Generic Mapping Tools: Improved version released, EOS Trans. AGU, 94, 409-410.

AdA temporary stations by the AdriaArray Seismology Group https://orfeus.readthedocs.io/en/latest/adria_array_main.html.

Permanent stations by national seismological services and ORFEUS-EIDA.

Big thanks to all the network operators for supplying the station information.

AdriaArray initiative is supported by Orfeus / EPOS.

petr.kolinsky@ig.cas.cz



discussion